

SZABIST Network – Master Config File

1. Global Device Security (Passwords)

Where to apply: Every single Router and Switch.

What it does: Encrypts all passwords and locks down the Console (physical cable), Telnet (remote access), and Privileged (`enable`) modes with the password `kkkr27`.

```
enable
configure terminal
service password-encryption
enable secret kkkr27
```

```
line console 0
password kkkr27
login
exit
```

```
line vty 0 4
password kkkr27
login
exit
```

2. Campus Switching (VLANs & Access Ports)

Where to apply: All Switches.

What it does: Creates the virtual departments (VLANs) and assigns specific PC ports to them. It also sets the connection between the switches and routers to "Trunk" mode so all VLANs can travel across one cable.

enable
configure terminal

! 1. Create the VLAN

```
vlan [VLAN_ID]  
name [DEPARTMENT_NAME]  
exit
```

! 2. Assign PC ports to the VLAN

```
interface range fastEthernet [START_PORT] - [END_PORT]  
switchport mode access  
switchport access vlan [VLAN_ID]  
exit
```

! 3. Configure the Uplink (Cable going to another Switch or Router)

```
interface [GIGABIT_PORT]  
switchport mode trunk  
exit
```

3. Campus Routing (Inter-VLAN & DHCP Helper)

Where to apply: All Campus Routers (100, 99, 154, 153, 78, Villas).

What it does: Divides the router's single physical cable into virtual "sub-interfaces" so it can act as the Default Gateway for every VLAN. It also tells the router to forward IP address requests to your EDGE router.

enable
configure terminal

! Enter the virtual sub-interface for a specific VLAN
interface gigabitEthernet 0/0.[VLAN_ID]
encapsulation dot1Q [VLAN_ID]
ip address [GATEWAY_IP] [SUBNET_MASK]

! Forward DHCP requests to the EDGE Router
ip helper-address 11.11.11.1
exit

4. The Brains (OSPF Dynamic Routing)

Where to apply: All Routers (including EDGE).

What it does: Activates a mathematical algorithm that allows all routers to automatically share their connected networks with each other, forming a unified campus backbone.

```
enable
configure terminal
router ospf 1
```

```
! Advertise a network directly connected to this router
network [NETWORK_IP] [WILDCARD_MASK] area 0
exit
```

5. The Edge Router (DHCP & NAT to Internet)

Where to apply: The `EDGE-RTR` only.

What it does: Hosts the central DHCP server that hands out IPs to every campus. It also translates all internal private IPs into your public IP (`11.11.11.1`) so students can browse the internet.

enable
configure terminal

! 1. Create a DHCP Pool for a specific department

```
ip dhcp pool [POOL_NAME]  
network [NETWORK_IP] [SUBNET_MASK]  
default-router [GATEWAY_IP]
```

```
dns-server 8.8.8.8
```

```
exit
```

! 2. Setup NAT Translation (Inside to Outside)

```
interface [PORT_FACING_ISP]  
ip nat outside  
exit  
interface [PORT_FACING_CAMPUS]  
ip nat inside  
exit
```

! 3. Activate NAT Overload

```
access-list 1 permit any  
ip nat inside source list 1 interface [PORT_FACING_ISP] overload
```

! 4. Send all unknown traffic to the Internet (Default Route)

```
ip route 0.0.0.0 0.0.0.0 11.11.11.2
```

6. Lab Security (Sticky MAC Port Security)

Where to apply: Sub-switches inside student labs (CS Lab, Mechatronics, etc.).

What it does: Memorizes the exact MAC address of the first PC plugged into the wall port. If a student unplugs the PC and plugs in their personal laptop, the port instantly drops the traffic.

```
enable
configure terminal
interface range fastEthernet [START_PORT] - [END_PORT]
```

! Turn on Port Security
switchport port-security

! Automatically memorize the PC's MAC address
switchport port-security mac-address sticky

! Drop traffic from unauthorized devices
switchport port-security violation restrict
exit

7. Network Firewalls (Access Control Lists - ACLs)

Where to apply: Main Campus Switches (via VLAN interfaces).

What it does: Blocks specific departments from talking to sensitive devices (like Faculty Printers or the ZABDESK server).

enable
configure terminal

! 1. Create the Security Rule

ip access-list extended [RULE_NAME]

! Permit authorized subnets to reach the target IP

permit ip [AUTHORIZED_NETWORK_IP] [WILDCARD_MASK] host
[TARGET_DEVICE_IP]

! Block everyone else from the target

deny ip any host [TARGET_DEVICE_IP]

! Allow all other normal internet/network traffic to pass

permit ip any any
exit

! 2. Apply the rule to the VLAN gateway

interface vlan [VLAN_ID_OF_THE_TARGET_DEVICE]
ip access-group [RULE_NAME] out
exit

 Pro-Tip for your Presentation:

If your instructor asks *why* you used certain technologies, keep these short answers in mind:

- **Why VLSM?** "To prevent wasting IP addresses. We gave smaller masks like /28 to small departments and /26 to big labs."
- **Why OSPF?** "Because if a cable breaks, OSPF automatically finds a backup path. Static routing requires manual fixing."
- **Why Port Security?** "To prevent unauthorized devices from entering our network and causing IP conflicts or security breaches."